



TYPE CM OIL-IMMERSED ON LOAD TAP CHANGER

Shanghai Huaming Power Equipment Co.,Ltd.



TYPE CM OIL-IMMERSED ON LOAD TAP CHANGER

CM on load tap changer comprises of diverter switch and tap selector. It's applicable to power transformer, rectification transformer and furnace transformer with highest equipment voltage up to 252kV, and max.rated through-current 600A for three phases and 1500A for single phase. Three-phase OLTC is used at neutral point regulation for star connection, and single-phase at any connections.

The regulation configuration of CM type on load tap changer can be linear regulation, reversing regulation and coarse/fine regulation. The max. step voltage is 3300V and max. number of operating position is 35 for general model and 107 for multiple coarse regulation.

Major structure properties

▶ Adopting parallel dual-contact structure, CM type OLTC has high breaking capacity. It is suitable for power transformers, and industrial transformers which with frequent overload.



▶ Adopting gunlock structure and double-press springs, its driving mechanism is reliable and stable. The mechanical life reaches one million operations proved by test.



▶ Adopting shape moving contacts, with its self-aligning function, the tap changer can carry high current load and has good thermal and dynamic stability.



▶ Replacing phenolic laminated paper board with pultruded epoxy fiberglass bar, bending strength of the tap selector insulation bars has increased by 8 times.



Major components of product

Head-flange, Oil compartment of diverter switch, Diverter switch, Tap selector



Head-flange

All aluminum alloy parts of head flange are made in low pressure die-casting process, which features in good appearance, compact crystal phase structure, high mechanical strength and free of leakage.

Diverter switch

A compression-molding supporting cylinder, made of SMC, is connected rigidly with upper plate and lower spring energy accumulator, thus ensuring reliable fixing and accurate positioning of contacts, and higher mechanical strength, which avoids decrease of main insulation due to damp and crack of phenolic paper board.

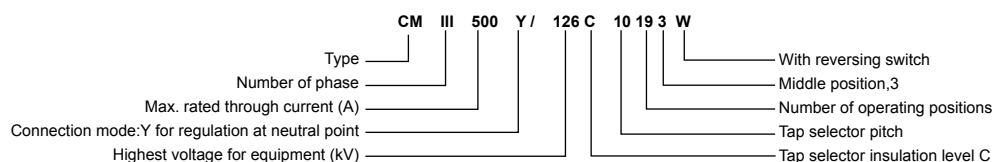
Oil compartment

Replacing paper laminated cylinder with wound epoxy fiberglass cylinder has improved electrical and mechanical strength of oil compartment.

Tap selector

By replacing phenolic laminated board with pultrusion fiberglass bar in tap selector, the insulation level and mechanical strength of on-load tap changer has considerably increased, furthermore, bending strength has increased by 8 times, which effectively avoids that stationary and moving contacts fail to make resulting from deformation of insulation bars.

Designation of CM Type





Technical data of CM series OLTC

Specification		CMIII500 CMI500	CMIII600 CMI600	CMI800	CMI1200	CMI1500
Max. rated through-current (A)		500	600	800	1200	1500
Rated frequency(Hz)		50/60				
Connection		Three-phase for neutral point of star connection Single-phase for any selectable winding connection				
Max. rated step voltage (V)		3300				
Rated step capacity (kVA)		1400	1500	2000	3100	3500
Short-circuit current test (kA)	Thermal (3s)	8	8	16	24	24
	Dynamic (peak)	20	20	40	60	60
Max. Operating Positions	Without change-over selector	18 (special linear regulation can be up to 34)				
	With change-over selector	35 (special multiple coarse regulation can be up to 107)				
Insulation to ground (kV)	Highest voltage for equipment Um	72.5	126	170	252	
	Rated separate source AC withstand voltage (kV/50Hz,1min)	140	230	325	460	
	Rated lightning impulse withstand voltage (kV,1.2/50μs)	350	550	750	1050	
Tap selector		4 grades of B, C, D and DE according to insulation level				
Mechanical life		Not less than 800,000 operations				
Contact life		Not less than 200,000 operations				
Oil compartment of diverter switch	Service pressure	0.03Mpa				
	Leakage test	No leakage under 0.08Mpa for 24 hours				
	Over pressure protection	Bursting cap bursts at 300±20%kPa				
	Protection relay	Oil flow speed set at 1.0m/s±10%				
Motor drive unit		SHM or CMA7				
Oil filter plant		ZXJY-1 /ZXJY-3 according to requirement				

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